

3 The photograph shows pea seeds with growing roots.



(Source: © Sangita Pal / Alamy Stock Photo)

Mitosis occurs in growing roots.

A scientist allowed pea seeds to start growing in different concentrations of oxygen.

Pea seeds were grown in oxygen concentrations of 5% and 20%.

After two days the root tips were removed.

The tips were stained and observed under a microscope.

The mitotic index for each root tip was determined.

The results

Mitotic index in 20% oxygen after two days

5.8 5.7 5.1 6.0 6.1 6.2 5.4 5.4 5.6

Mitotic index in 5% oxygen after two days

3.6 3.8 4.3 3.9 4.4 4.2 4.1 3.9 3.8

(a) State a suitable null hypothesis for this investigation.

(1)

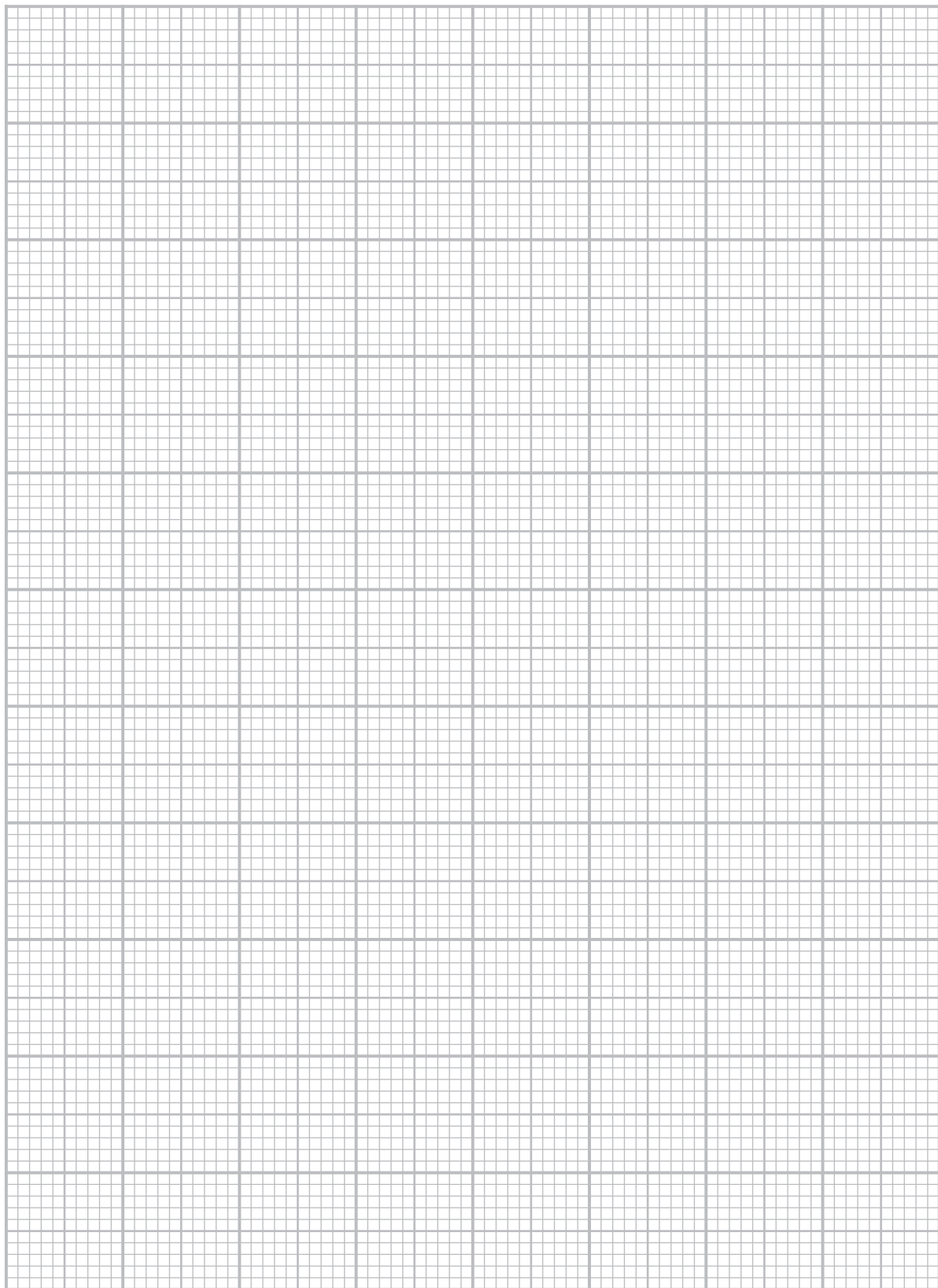
(b) Draw a suitable table to display the **data** and your calculated **means**.

(3)

- (c) Draw a suitable graph to show the mean mitotic index after two days in 20% oxygen and the mean mitotic index after two days in 5% oxygen.

Include an indication of the variability of the data.

(3)



(d) The scientist analysed the data with a t test using the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$ is the mean value for each treatment

n is the number of samples for each treatment

$(S_A)^2 = 0.1325$ and $(S_B)^2 = 0.0700$

(i) Calculate the value of t .

(2)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	p = 0.05	p = 0.01
9	2.26	3.25
10	2.23	3.17
11	2.20	3.11
12	2.18	3.06
13	2.16	3.01
14	2.15	2.97
15	2.13	2.94
16	2.12	2.92
17	2.11	2.90
18	2.10	2.88

Describe the conclusions that can be drawn from this investigation.

Use the information in your graph and the table to support your answer.

(3)

(e) Describe how the scientist could extend this investigation to collect more data to either support or reject the hypothesis.

(2)

(Total for Question 3 = 14 marks)
